

A Survey on Anomaly Detection in IoT and Cloud Computing Security

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Abstract—Cybersecurity in the Internet of Things (IoT) is crucial for ensuring the safety of various applications. The advent of Artificial Intelligence (AI) has enabled innovative security solutions across multiple layers of the IoT ecosystem. As IoT rapidly integrates with edge computing to reduce latency compared to cloud computing, the cloud continues to offer high-speed computation, extensive data analytics, and enhanced security. This survey examines anomaly detection in Cloud-IoT environments, focusing on maintaining safety and functionality. We analyze current security threats, such as data breaches and unauthorized access, while also discussing the benefits and challenges of Cloud IoT, including scalability and privacy concerns. This study aims to explore the effectiveness of cloud-based IoT deployments by reviewing cutting-edge methodologies.

Index Terms—Anomaly Detection, Internet of Things (IoT), IoT Security, Cloud Computing, Cloud Security, Machine Learning, Deep Learning, Edge Computing.

I. INTRODUCTION

The IoT architecture has evolved, and the Information and Communications Technology (ICT) standards development organization usually proposes the IoT framework. The three-layered IoT architecture consists of the network layer, perception layer, and application layer. Figure 1 depicts a more complex IoT architecture, with sensors and actuators at the device or perception layer, routers, internet gateways, and other communications components at the network layer, with edge computing being introduced the Edge IoT layer will have the edge processing and storage, and finally the cloud layer with its processing, storage and applications.

Various sensing, communication, and processing technologies are used to provide the necessary services. At each layer, analytics and management control is employed through the path of data flow. Security is a key feature that is required at each stage of the IoT architecture, it applies to device security, data security, network security, secure storage, secure processing, and secure analytics

The convergence of IoT and cloud computing has significantly transformed industries by enabling real-time data processing, storage, and analysis. IoT devices generate vast amounts of data, necessitating the efficient processing capabilities offered by cloud infrastructures. This integration

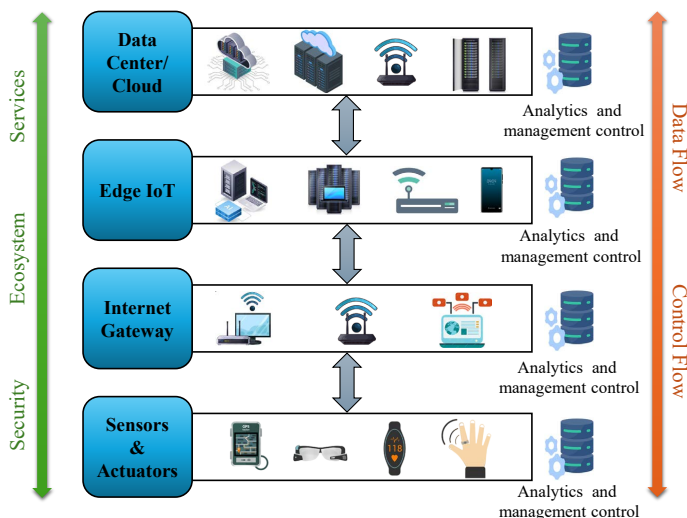


Fig. 1. Internet-of-Things Solutions Architecture.

supports applications in smart homes, healthcare, industrial automation, and intelligent transportation systems, thereby enhancing operational efficiency and decision-making processes [1]. However, this integration also introduces vulnerabilities that cyber threats can exploit, making anomaly detection essential for robust security strategies [2].

Determining patterns in data that differ from anticipated behavior is known as anomaly detection, and it is essential for stopping malicious activity and guaranteeing system dependability [3]. This paper provides a comprehensive survey of state-of-the-art techniques for anomaly detection in IoT and cloud computing, explores the challenges in this domain, and outlines future research directions. The goal is to identify key gaps in existing research and propose innovative solutions to enhance the security and robustness of IoT-cloud systems [4].

The paper is structured as follows: The Introduction is covered in Section I, the Survey of anomaly detection algorithms is described in Section II, and the IoT Cloud Security Challenges are discussed in Section III. Section IV presents the assessment metrics, while Section V explores future directions

and Section VI provides a conclusion.

II. RELATED WORK

Recent advancements in anomaly detection within IoT and cloud computing have focused on developing advanced models and techniques to address the unique challenges posed by these environments. This section reviews significant contributions from recent studies, providing a comprehensive overview of the current research literature. The paper covers the anomaly detection concept over various categories including, IoT, Deep Learning (DL), Machine Learning (ML), Cloud Computing, and Cybersecurity. The distribution of the percentage of papers referenced in each category is shown in the Figure 2

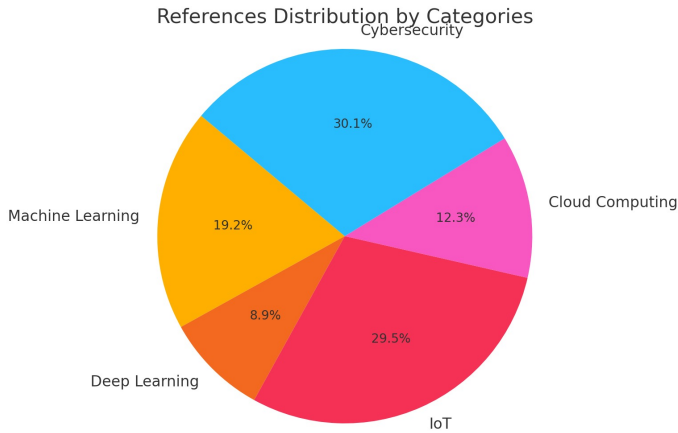


Fig. 2. Percentage distribution of papers referred for the survey.

A. Different Approaches for Anomaly Detection Techniques

1) *Machine Learning-Based Approaches:* ML approaches have become instrumental in detecting anomalies within IoT environments due to their ability to learn and adapt to dynamic data patterns. Research by Aznoli and Zarrin [5] explores the optimization of decision trees, support vector machines (SVM), and neural networks for IoT systems. These models can efficiently process large volumes of data with minimal latency, which is crucial given the resource constraints typical of IoT devices [6]. Furthermore, advancements in transfer learning have allowed ML models to adapt to new environments with minimal labeled data, thereby reducing the time and resources needed to train models from scratch [7]. Techniques such as online learning and incremental learning enable ML models to update their parameters in real-time, improving the models' responsiveness to evolving threats [8].

Globally, researchers are exploring the use of ML algorithms for predictive maintenance in industrial IoT applications, where anomalies in sensor data can indicate potential equipment failures [9]. In smart city deployments, ML models are employed to monitor traffic patterns and detect irregularities that may signify accidents or congestion [10]. By integrating cloud computing's scalability, these models can handle the high data throughput typical of urban environments [11].

2) *Deep Learning Enhancements:* DL models have shown remarkable capabilities in extracting high-level features from complex datasets, making them suitable for anomaly detection in IoT environments [12]. Research by Kim and Ko [13] highlights how Convolutional Neural Networks (CNNs) and long short-term memory (LSTM) networks can process large-scale IoT data streams, identifying anomalies with improved accuracy and speed. CNNs are particularly effective for spatial data analysis, such as image-based IoT applications, while LSTMs excel in analyzing temporal sequences like time-series data from sensors [14].

The integration of DL models with edge computing further enhances their utility in IoT systems [15]. By processing data locally at the edge, DL models can reduce latency and bandwidth usage, enabling real-time anomaly detection in critical applications such as autonomous vehicles and healthcare monitoring [16]. Recent studies have also explored the use of generative adversarial networks (GANs) to simulate adversarial attacks, allowing DL models to be trained for improved resilience against such threats [17].

On a global scale, DL approaches are being applied to enhance security in smart grid systems by detecting irregular power consumption patterns that may indicate cyber intrusions [18]. In the financial sector, DL models are used to monitor transaction data for fraud detection, integrating cloud computing to scale across millions of transactions per second [19].

3) *Hybrid Techniques:* Hybrid techniques that combine multiple methodologies offer robust solutions to the complex challenges of anomaly detection in IoT environments. These techniques integrate statistical, machine learning, and deep learning approaches to utilize their complementary strengths, enhancing detection accuracy and robustness [20]. For instance, hybrid models that combine statistical anomaly detection with deep learning can better handle diverse data types and improve detection reliability in heterogeneous IoT networks [21].

In the context of industrial IoT, hybrid approaches are used to balance the trade-offs between computational efficiency and detection performance. By combining lightweight statistical models with deep learning, systems can achieve high accuracy without overwhelming limited resources [22]. Recent advancements have explored the integration of hybrid models with blockchain technology to ensure data integrity and secure anomaly detection logs [23].

Globally, hybrid anomaly detection systems are being deployed in smart city projects, where they help manage and analyze data from a wide array of sensors to optimize urban infrastructure [24]. In healthcare, hybrid models enable precise monitoring of patient vitals and early detection of potential health issues [25].

4) *Distributed Anomaly Detection:* Distributed anomaly detection approaches integrate the computational power of both cloud and edge devices to improve real-time detection capabilities and reduce latency [26]. By distributing the computational load across multiple nodes, these methods ensure scalable processing of IoT data [27]. Due to its ability to

facilitate local processing and minimize the quantity of data that must be sent to the cloud, edge computing is essential to distributed detection [28].

Recent research highlights the effectiveness of distributed frameworks in handling large-scale IoT deployments, such as those in smart agriculture and environmental monitoring [29]. These systems can dynamically allocate resources and adapt to changing data patterns, ensuring efficient anomaly detection in real-time [30].

Distributed anomaly detection is also being applied in the financial sector to monitor transactions across decentralized networks, improving the detection of fraudulent activities while maintaining data privacy [31]. In the automotive industry, distributed systems support the real-time analysis of vehicle data for predictive maintenance and accident prevention [32].

5) Privacy-Preserving Methods: Privacy-preserving anomaly detection techniques have gained attention, particularly in sensitive applications such as healthcare and smart homes [33]. These methods aim to detect anomalies while ensuring the confidentiality of user data, often using techniques like homomorphic encryption and differential privacy [34]. Strong privacy protection is provided by homomorphic encryption, which increases computational costs by enabling calculations on encrypted data without first decrypting it. [35].

Differential privacy, on the other hand, introduces noise to the data or its outputs, making it difficult to infer individual data points from aggregated results [36]. This approach is particularly valuable in environments where user data is highly sensitive, such as medical records and financial transactions [37].

Globally, privacy-preserving methods are being adopted in smart city projects to protect citizen data while enabling collaborative anomaly detection across various departments [38]. In the education sector, these techniques ensure the privacy of student data while analyzing patterns in digital learning platforms [39].

6) Federated Learning Approaches: Federated learning is a viable paradigm for collaborative anomaly detection that enables the training of shared models across different devices without requiring the exchange of raw data [40]. This approach preserves privacy and minimizes communication overhead by only transmitting model updates rather than sensitive data [41]. Federated learning is particularly beneficial in IoT environments, where bandwidth and privacy are major concerns [34].

Federated learning has been shown to be beneficial in healthcare applications in recent studies, where it enables the collaborative analysis of patient data across hospitals without compromising patient privacy [42]. In smart grid systems, federated learning facilitates the detection of anomalies in power consumption patterns while maintaining the confidentiality of user data [43].

The automotive industry has also begun to implement federated learning for anomaly detection in vehicle systems, allowing manufacturers to improve the safety and reliability of connected cars while safeguarding user data [44]. Similarly,

federated learning is being explored in the financial sector to enhance fraud detection across decentralized transaction networks [45].

7) Adversarial Attack Resilience: Recent research has focused on enhancing the resilience of anomaly detection systems against adversarial attacks, which attempt to manipulate data to evade detection [46]. Techniques such as adversarial training and robust feature extraction are being explored to improve system robustness [47]. Adversarial training involves augmenting training datasets with adversarial examples, which helps models learn to recognize and mitigate adversarial manipulations [48].

Robust feature extraction methods aim to identify features that are less susceptible to adversarial perturbations, enhancing the resilience of anomaly detection models [49]. These techniques are particularly important in IoT environments, where adversarial attacks can exploit the limited resources and computational capabilities of devices [50].

Globally, adversarial attack resilience is being incorporated into smart city security frameworks to protect critical infrastructure from cyber threats [51]. In the healthcare industry, robust anomaly detection models are used to safeguard patient data from adversarial attacks while ensuring the accuracy of medical diagnoses [52]. The financial sector also relies on adversarial training to protect transaction data from manipulation and fraud [53].

B. Anomaly Detection Techniques for IoT and Cloud Environment

Anomaly detection is employed in data privacy and security in the IoT. Currently, ML and AI are the technologies extensively used to propose novel security solutions. For understanding the application of security solutions to IoT it is important to understand the taxonomy of IoT security [54]. The key elements of the IoT security primitives are shown in Figure 3.

Taxonomy of Security in IoT			
Data	Communication	Architecture	Application
- Privacy - Trust	- Information Exchange - Information Sharing	- Authentication - Authorization	- Authentication - Authorization - Resource Sharing - Trust Establishment

Fig. 3. Taxonomy of IoT security.

Anomaly detection technology is employed for security of data, communication, secure authentication, and authorization in various IoT applications based on edge computing and cloud computing architectures [55]. Various ML algorithms are used for anomaly and intrusion detection to provide security of cyber-physical systems of the IoT, suitable for the edge layer. Edge computing demands lightweight computational models and studies have shown that many such ML models are suitable for anomaly detection at the edge [56].

Anomaly detection techniques in IoT and cloud environments can be categorized into statistical methods, machine learning approaches, deep learning models, and hybrid techniques. Each category offers unique strengths and challenges, making them suitable for different scenarios.

1) *Statistical Methods*: Statistical methods provide foundational techniques for anomaly detection, relying on probabilistic models to identify deviations from expected patterns [3]. Gaussian models, Bayesian networks, and hypothesis testing are commonly used approaches in this category.

•Gaussian Models

Gaussian models assume that data follows a normal distribution, flagging deviations that fall outside a predefined confidence interval as anomalies [57]. However, Gaussian models may struggle with high-dimensional or non-normally distributed data typical of IoT environments [58]. They are effective in applications where data distributions are well understood and can be modeled accurately [59].

•Bayesian Networks

Bayesian networks illustrate a set of variables and their conditional relationships using probabilistic graphical models [60]. These networks are advantageous in scenarios where data uncertainty is prevalent, as they allow for robust anomaly detection in dynamic IoT systems [61]. They provide a flexible framework for incorporating prior knowledge and reasoning under uncertainty [62].

•Hypothesis Testing

Hypothesis testing involves the formulation of null and alternative hypotheses about data behavior, followed by statistical testing to identify significant anomalies [63]. This method is useful for applications requiring rigorous statistical validation of anomalies. Common tests include t-tests, chi-square tests, and likelihood ratio tests, which help determine whether observed anomalies are statistically significant [64].

2) *Machine Learning Approaches*: ML techniques have been extensively adopted for their flexibility and capability to model complex data relationships [65]. These approaches include supervised, unsupervised, and semi-supervised learning methods.

•Supervised Learning

Supervised learning techniques such as decision trees and SVMs require labeled datasets for training [66]. These models are effective in environments where historical data accurately captures both normal and anomalous behavior [67]. However, obtaining labeled data can be challenging in dynamic IoT environments. Transfer learning and data augmentation techniques are often employed to mitigate the scarcity of labeled data [7].

•Unsupervised Learning

Unsupervised learning methods, including clustering algorithms like k-means and DBSCAN, do not require labeled data and are effective in detecting novel anomalies that differ significantly from normal data patterns [68]. These methods excel in identifying outliers in datasets where the definition of normal behavior is not well-defined [69].

•Semi-supervised Learning

Semi-supervised learning techniques utilize a combination of labeled and unlabeled data, providing a balance between the advantages of supervised and unsupervised learning [70]. These methods are particularly useful in scenarios where labeled data is scarce or costly to obtain. Techniques like self-training, co-training, and graph-based methods utilize the limited labeled data to improve detection accuracy [71].

3) *Deep Learning Models*: Large-scale IoT data can be handled by DL models as they can automatically extract complex characteristics from raw data, which has made them popular [14].

•Convolutional Neural Networks (CNNs)

CNNs are effective for spatial data analysis, such as image-based IoT data. They are designed to recognize hierarchical structures within data, making them ideal for detecting spatial anomalies [72]. CNNs have been applied to various IoT applications, including image recognition in smart cameras and anomaly detection in industrial automation [73].

•Recurrent Neural Networks (RNNs)

RNNs are well-suited for sequential data analysis, such as time-series data generated by IoT sensors. Their ability to maintain temporal context makes them powerful for detecting temporal anomalies [74]. Variants like LSTMs and GRUs address the vanishing gradient problem, allowing them to capture long-term dependencies in sequential data [75].

•Autoencoders

Autoencoders are neural networks that are able to learn compressed representations of data, recreate input data, and identify deviations as abnormal. [76]. Autoencoders are particularly effective for unsupervised anomaly detection in high-dimensional datasets. They have been used in applications such as network traffic analysis and fault detection in smart grids [77].

4) *Hybrid Methods*: Hybrid methods integrate multiple techniques to utilize their complementary strengths, improving detection accuracy and robustness [78].

•Statistical and Machine Learning Integration

Combining statistical models with ML algorithms enhances anomaly detection by providing a more comprehensive understanding of data distributions and patterns [79]. These approaches can improve detection accuracy by introducing statistical insights to guide machine learning model training [80].

•Deep Learning and Traditional Approaches

Integrating DL models with traditional statistical methods enables the handling of complex data patterns while retaining the interpretability of simpler models [81]. Hybrid models can capture both linear and nonlinear relationships, providing an overview of data anomalies. [57].

•Adaptive Hybrid Systems

Adaptive hybrid systems dynamically adjust their detection strategies based on the evolving characteristics of IoT data, ensuring continuous adaptation to changing environments [82]. Adaptive systems employ reinforcement learning and online learning techniques to update models in real-time, accommodating new patterns and behaviors [83].

TABLE I
SUMMARY OF RECENT ANOMALY DETECTION ALGORITHM

Author(s) (Year)	Methodology Used	Advantages	Disadvantages
S. G. Aarela et al. (2023) [55]	Machine Learning, Edge Computing	Enhances security, reduces latency in edge devices	Implementation complexity and high setup costs
P. K. Sadhu et al. (2022) [54]	IoT Security, Machine Learning	Comprehensive framework for IoT security	Theoretical, requires real-world validation
E. Altunhan (2024) [24]	Smart City Security, Privacy	Solutions for smart city challenges	Requires large-scale infrastructure
O. Olateju et al. (2024) [12]	Anomaly Detection in Cyber-Physical Systems	Effective for real-time anomaly detection	High computational requirements for large systems
H. Li et al. (2024) [13]	Deep Learning, IoT, Edge Computing	Reduces latency and improves bandwidth usage	High resource demands at the edge
T. A. Devi et al. (2024) [14]	Deep Learning	High accuracy in processing complex datasets	Requires vast computational resources and large datasets
L. Hasini et al. (2024) [16]	Edge Computing, AI	Optimizes real-time processing in distributed systems	Difficult to integrate with existing infrastructure
D. Dhinakaran et al. (2024) [20]	Hybrid Anomaly Detection	Combines machine learning and deep learning for high accuracy	Resource-intensive, complex to maintain
Nguyen et al. (2021) [34]	Federated Learning in IoT	Preserves privacy while minimizing communication overhead	Complex coordination among distributed devices
J. Ramasam et al. (2024) [29]	Privacy-Preserving Data Mining	Protects sensitive data during analysis	Computational overhead due to encryption
A. Ravuri et al. (2024) [27]	Fog Networks, Energy Efficiency	Optimizes energy consumption in distributed IoT environments	Limited application in large-scale systems
P. Reddy et al. (2024) [28]	Edge Computing	Reduces latency for real-time IoT applications	Security vulnerabilities and resource constraints
A. Nag et al. (2024) [25]	Machine Learning in Healthcare	Effective in detecting anomalies in medical datasets	Data privacy and limited data access concerns
K. Mitropoulou et al. (2024) [19]	Machine Learning for Financial Applications	Enhanced fraud detection and risk analysis	Vulnerable to adversarial attacks on models
P. Raghunwansi (2024) [23]	Blockchain and IoT Integration	Enhances security and integrity of IoT data	Scalability issues in large distributed networks
A. D. Vibhute et al. (2024) [18]	Smart Grid Anomaly Detection	Effective in real-time monitoring of smart grids	High complexity in integrating with existing grid infrastructure
R. Gupta et al. (2020) [84]	Network Anomaly Detection in IoT	Reduces false positives in IoT anomaly detection	Requires extensive labeled data for training
M. Zaharia et al. (2020) [85]	Apache Spark for Big Data Processing	High scalability and performance for big data analytics	Requires significant resources for distributed systems
Y. Cui et al. (2020) [86]	Adaptive Sampling for Big Data	Reduces data transmission, improves system efficiency	Limited applicability for all types of data streams
T. Wang et al. (2020) [32]	Distributed Anomaly Detection	Efficient in large-scale IoT networks	High coordination complexity in distributed environments
S. Han et al. (2020) [87]	Deep Compression for Neural Networks	Reduces memory and computation requirements	Can negatively affect model accuracy with aggressive compression
J. Zhao et al. (2020) [33]	Privacy-Preserving Machine Learning	Ensures data privacy without compromising performance	Requires complex cryptographic protocols
C. Molnar (2020) [88]	Interpretable Machine Learning	Enhances transparency and interpretability of ML models	Limited to simpler models, lacks in-depth understanding of complex DL
N. Akhtar & A. Mian (2020) [89]	Adversarial Attacks on Deep Learning	Identifies vulnerabilities and improves system robustness	Requires complex countermeasures, difficult to mitigate fully
S. Kanhere et al. (2020) [90]	Community-Driven Anomaly Detection in IoT	Engages community participation for more efficient anomaly detection	Data privacy and security challenges in community-shared data
D. Vikram et al. (2020) [17]	Generative Adversarial Networks (GANs)	Simulates adversarial attacks for model robustness testing	Complex to train, susceptible to mode collapse
K. Rezaei et al. (2024) [10]	IoT Applications for Smart Cities	Enhances management of urban infrastructure with IoT systems	Privacy and security challenges in large-scale implementations
M. Ribeiro et al. (2020) [91]	Explainable Machine Learning	Enhances user trust by explaining model decisions	Reduces complexity in model explanations

The Table I summarizes the discussions on the various methodologies employed for anomaly detection in security of cloud and IoT environments.

III. CHALLENGES IN IoT AND CLOUD SECURITY

Despite the advancements in anomaly detection techniques, several challenges persist in ensuring robust security in IoT and cloud environments. This section discusses key challenges and potential solutions to address them.

A. Data Privacy

Data privacy is a significant concern in IoT and cloud environments due to the sensitive nature of data being processed and stored [36]. Protecting user privacy while ensuring effective anomaly detection remains a critical challenge.

1) *Federated Learning*: With this method, data is kept locally, protecting privacy, and IoT devices may work together to jointly train a shared anomaly detection model. To ensure that raw data stays on the devices, model changes are pooled on the cloud. Without sacrificing data security, federated learning allows for scalability model training across dispersed devices [34].

2) *Differential Privacy*: In order to keep sensitive information hidden, differentiating privacy strategies introduce noise into data or model changes [92]. This guarantees the outputs of anomaly detection algorithms do not allow for the inference of individual data points. Differential privacy provides strong theoretical guarantees of privacy protection, making it suitable for applications where data confidentiality is paramount [93].

3) *Homomorphic Encryption*: Homomorphic encryption maintains privacy and makes anomaly detection possible in cloud environments by allowing calculations to be done on encrypted data without decrypting it [35]. This approach can be computationally expensive but offers robust privacy protection for sensitive data [94].

B. Scalability

The high amount and velocity of data generated by IoT devices makes scalability crucial in IoT-cloud systems [95]. Ensuring that anomaly detection systems can scale efficiently to handle this data is a major challenge.

1) *Cloud-Native Solutions*: The integration of cloud-native architectures allows for elastic scaling of resources, accommodating fluctuating data loads without compromising performance [96]. Cloud-native solutions provide the infrastructure necessary to process and store vast amounts of IoT data, ensuring that detection systems remain responsive under varying conditions [97].

2) *Distributed Computing Frameworks*: Distributed frameworks such as Apache Kafka and Apache Flink provide scalable data processing capabilities, enabling real-time anomaly detection across large IoT deployments [98]. These frameworks support the distribution of computational load across multiple nodes, improving scalability and reducing latency [85].

3) *Edge Computing*: Offloading data processing to edge devices reduces latency and bandwidth usage, allowing for real-time detection close to data sources [15]. Edge computing can alleviate the burden on cloud resources and improve response times for time-sensitive applications [28].

C. Resource Constraints

IoT devices often operate under resource constraints, limiting their ability to execute complex anomaly detection algorithms [99]. Developing efficient methods that account for these limitations is crucial for effective security.

1) *Lightweight Models*: Developing lightweight models for resource-constrained devices ensures efficient anomaly detection without overburdening device capabilities [75]. Strategies such as quantization and model pruning can decrease the computational requirements of DL models [87].

2) *Adaptive Sampling Techniques*: By reducing the volume of data handled by anomaly detection systems, adaptive sampling approaches can save resources while preserving detection accuracy [86]. These techniques dynamically adjust the sampling rate based on data characteristics and system requirements [100].

3) *Energy-Efficient Algorithms*: Designing energy-efficient algorithms that minimize power consumption while maintaining detection performance is essential for battery-powered IoT devices [26]. Low-power communication protocols and duty cycling are two strategies that can increase the operating life of Internet of Things devices [101].

D. Adversarial Attacks

Anomaly detection systems are vulnerable to adversarial attacks that aim to manipulate data and evade detection [89]. Developing robust techniques to defend against such attacks is critical for ensuring system security.

1) *Adversarial Training*: Adversarial training involves adding adversarial samples to augmenting training data in order to strengthen the adaptability of detection algorithms [48]. This approach helps models learn to recognize and mitigate adversarial manipulations [46].

2) *Robust Feature Extraction*: Implementing robust feature extraction techniques can enhance the resilience of anomaly detection systems by focusing on features that are less susceptible to adversarial perturbations [49]. These techniques improve the model's ability to generalize across different attack scenarios [47].

3) *Anomaly Explanation Models*: Developing models that can explain detected anomalies can improve system transparency and facilitate the identification of adversarial attacks [102]. Explanation models provide insights into the reasoning behind anomaly detection decisions, enabling users to assess the reliability of detection results [103].

IV. EVALUATION METRICS

Evaluation metrics are crucial for assessing the effectiveness of anomaly detection systems in IoT and cloud computing environments. This section outlines key metrics used to evaluate performance, efficiency, and robustness.

A. Accuracy

One of the key performance indicators for anomaly detection systems is accuracy. Precision, recall, and the F1-score are commonly used to quantify it since they collectively paint a complete picture of the system's capacity to differentiate between normal and abnormal input [104].

1) *Precision*: The precision measures the proportion of accurately detected anomalies to all defects detected by the system. Good precision results to a low percentage of false positives. [104].

2) *Recall*: The proportion of successfully identified anomalies to all actual abnormalities in the dataset is known as recall. A low rate of false negatives is indicated by high recall [104].

3) *F1-Score*: The F1-score is a balanced assessment statistic for detection accuracy, calculated as the harmonic mean of precision and recall. With a high F1-score, recall and precision are well-balanced, reducing false positives and false negatives [104].

B. Efficiency

Efficiency is critical for real-time anomaly detection, where quick response times are essential. Evaluation metrics for efficiency focus on system latency and throughput [105].

1) *Latency*: Latency is the time taken for the system to detect and respond to an anomaly. Low latency is crucial for timely threat mitigation, particularly in applications where immediate action is required [99].

2) *Throughput*: Throughput is the volume of data that can be processed by the system in a given time frame. High throughput indicates the system's ability to handle large data volumes, ensuring scalability and responsiveness [85].

3) *Resource Utilization*: The efficiency of resource usage, including CPU, memory, and network bandwidth, is critical for evaluating the practicality of anomaly detection systems in resource-constrained environments [28].

C. Robustness

Robustness measures the system's resilience to various types of anomalies and attacks. Evaluation metrics for robustness focus on the system's ability to generalize across different conditions and withstand adversarial attempts [48].

1) *Adversarial Robustness*: The system's ability to withstand adversarial attacks that attempt to evade detection by introducing subtle perturbations in data. Metrics for adversarial robustness assess the model's performance under attack scenarios and its ability to detect manipulated data [51].

2) *Generalization*: The system's capability to detect novel anomalies across different data distributions and conditions, ensuring its effectiveness in diverse environments. Metrics for generalization evaluate the model's adaptability to new and unseen data patterns [14].

V. LIMITATIONS OF EXISTING WORKS AND RESEARCH GAPS

Despite advances in anomaly detection, several limitations persist in existing methods. Statistical models tend to oversimplify data and struggle with high-dimensional datasets, often leading to false positives. Machine learning models, although powerful, require large amounts of labeled data, which can be difficult to acquire in real-world IoT applications. Moreover, they are resource-heavy, making them impractical for edge computing scenarios. Deep learning techniques offer improved accuracy but are computationally expensive and often unsuitable for low-power IoT devices. These limitations highlight a critical gap: the need for lightweight, real-time anomaly detection solutions that can function effectively in resource-constrained environments while maintaining high accuracy. This study proposes to address this gap by exploring hybrid techniques that balance scalability, efficiency, and precision across different IoT ecosystems.

1) *Robustness to Noise*: The system's resilience to noise and variations in input data, which can affect detection accuracy. Metrics for robustness to noise assess the system's ability to maintain performance in the presence of data uncertainties [57].

VI. FUTURE DIRECTIONS

The future of anomaly detection in IoT and cloud computing lies in addressing the challenges identified and the emerging technologies can be used to enhance the detection capabilities. Some of the future research and developments are discussed below.

A. Advanced Machine Learning Techniques

Emerging machine learning techniques offer promising avenues for improving anomaly detection systems by enhancing their accuracy, efficiency, and adaptability [14].

1) *Transfer Learning*: Transfer learning can improve anomaly detection by gaining knowledge from related domains, reducing the need for extensive labeled datasets [7]. This approach enables models to adapt quickly to new environments and data patterns, enhancing detection performance.

2) *Reinforcement Learning*: Applying reinforcement learning to anomaly detection allows systems to adapt to evolving threats and optimize detection strategies based on feedback. Reinforcement learning can enhance model adaptability and decision-making in dynamic environments.

3) *Graph-Based Learning*: Graph-based learning techniques can improve anomaly detection by capturing complex relationships and dependencies between data points. These techniques are particularly effective in IoT environments, where devices and data are often interconnected.

B. Integration with Blockchain

Blockchain technology offers unique capabilities for enhancing the security and transparency of anomaly detection systems in IoT and cloud environments.

1) *Decentralized Anomaly Detection*: Integrating blockchain with anomaly detection systems can enhance data integrity and provide a tamper-proof log of detected anomalies, improving transparency and trust. Blockchain's decentralized nature ensures that anomaly detection records are immutable and verifiable.

2) *Smart Contracts for Automated Response*: Smart contracts can automate responses to detected anomalies, enabling rapid mitigation and reducing the reliance on manual intervention. Automated responses can improve the efficiency and consistency of threat mitigation efforts.

3) *Blockchain-Based Privacy Solutions*: Blockchain technology can enhance privacy preservation in anomaly detection systems by enabling secure data sharing and access control. Privacy-preserving blockchain solutions can facilitate collaborative anomaly detection without compromising data confidentiality [23].

C. Collaborative Anomaly Detection

Collaborative approaches to anomaly detection can enhance system effectiveness by utilizing shared knowledge and resources across multiple domains and organizations [84].

1) *Cross-Domain Collaboration*: Encouraging collaboration between different IoT and cloud systems can improve anomaly detection by sharing threat intelligence and utilizing diverse data sources [106]. Cross-domain collaboration can enhance detection accuracy by providing a more comprehensive view of potential threats [107].

2) *Federated Anomaly Detection*: Federated learning can be extended to enable collaborative anomaly detection across multiple organizations while preserving data privacy [40]. Federated anomaly detection facilitates the development of shared models that benefit from collective data insights [34].

3) *Community-Driven Anomaly Detection*: Community-driven initiatives can enable the development of open-source anomaly detection tools and frameworks, promoting innovation and knowledge sharing [108]. Community engagement can drive the creation of robust and adaptable detection solutions [90].

D. Explainable AI for Anomaly Detection

Explainable AI (XAI) techniques can enhance the transparency and interpretability of anomaly detection systems, facilitating trust and user acceptance [109].

1) *Model Interpretability*: Developing interpretable models that provide insights into anomaly detection decisions can improve user understanding and trust [110]. Techniques such as feature importance analysis and decision tree visualization can enhance model interpretability [111].

2) *Anomaly Explanation*: Providing explanations for detected anomalies can help users assess the reliability of detection results and identify potential false positives [88]. Explanation models can offer insights into the factors contributing to anomaly detection decisions, improving system transparency [112].

3) *User-Centric Anomaly Detection*: Designing user-centric anomaly detection systems that consider user feedback and preferences can enhance system usability and acceptance [113]. User-centric approaches can develop detection strategies to align with user needs and expectations [114].

VII. CONCLUSION

In conclusion, while substantial progress has been made in anomaly detection for IoT and cloud computing, challenges remain, particularly in terms of real-time capabilities and computational efficiency. Shortages include the lack of scalability in resource-constrained environments and the vulnerability of models to adversarial attacks. Challenges lie in the need to balance accuracy with resource consumption, particularly in edge computing scenarios, where both bandwidth and power are limited. However, there are significant opportunities for future research.

Emerging technologies such as federated learning offer promising solutions for privacy-preserving anomaly detection without sacrificing performance. Additionally, the integration of blockchain can enhance security by providing decentralized, tamper-resistant records of anomaly detection logs. The convergence of these technologies with more adaptive machine learning techniques could enable more robust, scalable, and efficient anomaly detection systems for the IoT-cloud ecosystem.

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